

ADITYA NARAYAN RAY

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Technical Skills

Programming Languages: C++, Python, JavaScript, TypeScript, C
Backend & Systems: FastAPI, Node.js, Supabase, REST APIs
Frontend Technologies: React.js, Tailwind CSS, WebSockets, WebRTC, Three.js
Databases and Storage: PostgreSQL, MySQL, MongoDB, Google Drive API
Tools & Practices: Git, GitHub, Docker

Professional Experience

Open Source Contributor

Oct 2025 – Feb 2026

Jitsi

Remote

- Contributed **7 merged pull requests** across **3 repositories** (jitsi-meet, lib-jitsi-meet, jitsi-meet-electron-sdk) in a **production WebRTC** platform used by millions globally.
- Developed a **PTZ (Pan-Tilt-Zoom) camera control** feature via the **WebRTC constraints API**, integrating **TypeScript** UI components and resolving a critical **uninitialized field bug** in JitsiLocalTrack.

Summer Research Intern

May 2026 – Jun 2026

NISER (National Institute of Science Education and Research)

Bhubaneswar, Odisha

- Benchmarked **Weighted Nonlinear Least Squares (WNLLS)**, **Gaussian Process regression**, and **Physics-Informed Neural Networks (PINNs)** for estimating photon mass attenuation coefficients from **Cs-137 gamma-ray transmission data**; co-authoring a comparative methodology paper on the tradeoffs.
- Built and iteratively debugged PINN training pipelines across **five materials** (Al, Cu, Brass, Steel, Pb)

Education

Manipal University Jaipur

Aug 2024 – May 2028

Bachelor of Technology in Computer Science and Engineering

Jaipur, India

- CGPA: **9.82/10.0** — **Dean's List** Recipient for **4 Semesters** — **9.67, 10, 9.72, 9.90**
- Relevant Coursework: **Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Design and Analysis of Algorithms**

Technical Projects

Ecolympics – Waste Warriors NGO Portal | React.js, Node.js, Supabase, Google Drive API

Sep 2026

- Developed an RBAC-enabled dashboard during the **JPMorgan Chase Code for Good Hackathon**, successfully migrating the NGO from manual Google Docs tracking to a centralized evaluation portal.
- Architected a zero-cost storage pipeline by utilizing India's free NGO Google Workspace tier, storing raw files in **Google Drive** while managing metadata and reference links efficiently in **Supabase**.

Sentinel AI – NGO Rescue Coordination Platform | React, FastAPI, PostgreSQL, Docker

Aug 2026

- Built a full-stack platform consolidating citizen reports, FIRs, images, and voice notes into a unified case workflow, with **JWT-based role access control** across citizen, NGO admin, police, and volunteer roles.
- Designed a modular backend exposing **30+ REST endpoints**, integrating **Tesseract OCR, OpenAI Whisper (speech-to-text)**, and **Llama-3** for AI-assisted case risk prioritization and a rule-based shelter engine.

Do LLMs know when they don't know | Python, Llama, Qwen

Oct 2026

- Architected an experimental evaluation pipeline testing 160 benchmark questions across 4 conditions, processing 640 responses to quantify LLM epistemic overconfidence.
- Engineered a Logprob Probe Gate intervention that slashed hallucination rates by **89%** (from 57.5% down to 6.2%) and increased overall accuracy to **78.8%**.
- Implemented explicit abstention prompting, achieving a statistically significant (**McNemar's test, p=0.002**) hallucination reduction while maintaining a **0.0% false abstention rate** on verifiable facts.

Honors, Awards & Certifications

Problem Solving: LeetCode Rating: **1510**

Competition Awards: Secured **1st Place** out of **100+ competing engineering teams** at a Bug Fixing Competition, Evoque 2025. Finalist at JPMorgan Chase Code for Good Hackathon.

Academic Honors: Awarded **Dean's List** status for maintaining a **top 1% academic ranking** across **4 consecutive semesters**

Certifications: **Machine Learning Specialization** (Andrew Ng / DeepLearning.AI), **Microsoft Azure AI Fundamentals**

Leadership: **Web Development Head**, ACM MUJ SIGAI – lead and mentor a team of **40+ students**; **Buddy Mentor** for undergraduates